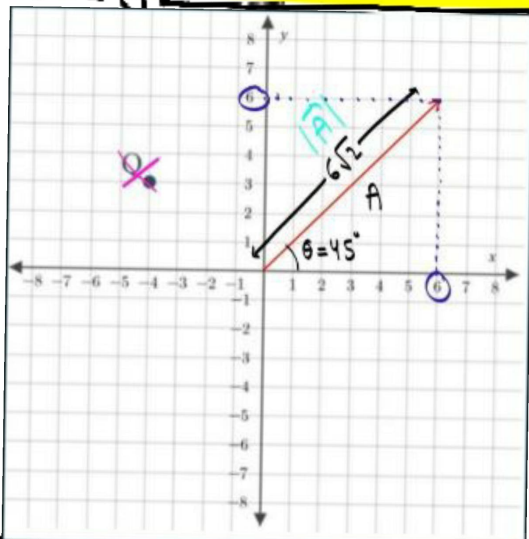


Motion In A Plane

Resolving a vector in 2D Plane:



x-component:

$$x = |\vec{A}| \cos \theta$$

y-component:

$$y = |\vec{A}| \sin \theta$$

$$\vec{A} = x\hat{i} + y\hat{j}$$

Find x, y ?

$$|\vec{A}| = 6\sqrt{2} \text{ given}$$

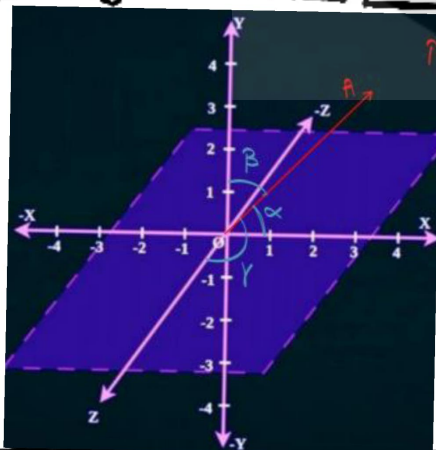
$$\theta = 45^\circ$$

Soln: $x = 6\sqrt{2} \cdot \cos(45^\circ)$

$$x = 6\sqrt{2} \times \frac{1}{\sqrt{2}} = x = 6$$

Similarly $y = 6$

Resolving a vector in 3D Plane:



x-component

$$x = |\vec{A}| \cos(\alpha)$$

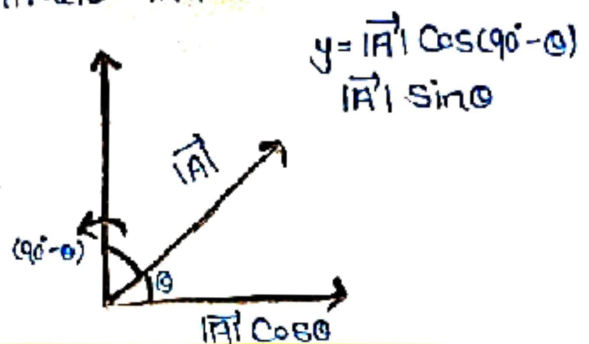
y-component

$$y = |\vec{A}| \cos(\beta)$$

z-component

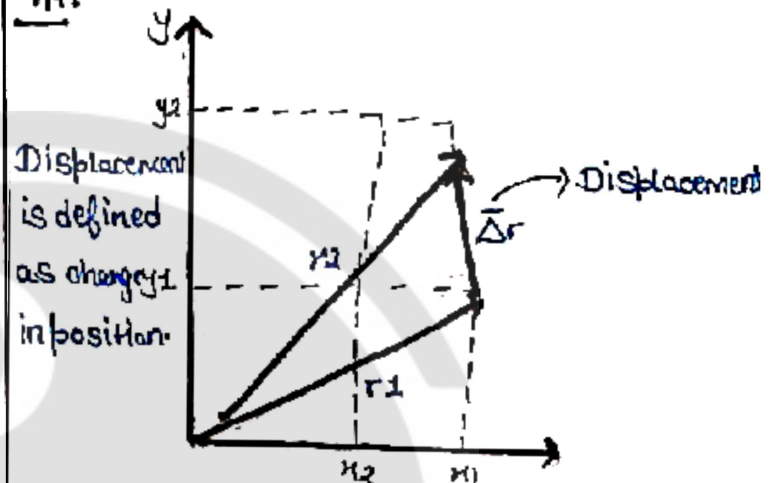
$$z = |\vec{A}| \cos \gamma$$

Magnitude = $|\vec{A}|$



Vectorial representation of Displacement

nt:



$$\vec{r}_1 = x_1\hat{i} + y_1\hat{j}$$

$$\vec{r}_2 = x_2\hat{i} + y_2\hat{j}$$

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\Delta \vec{r} = (x_2\hat{i} + y_2\hat{j}) - (x_1\hat{i} + y_1\hat{j})$$

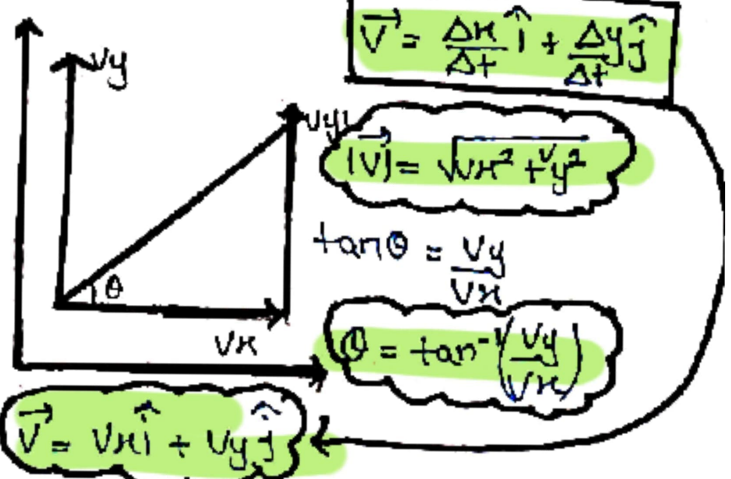
$$\Delta \vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}$$

$$\Delta \vec{r} = \Delta x\hat{i} + \Delta y\hat{j}$$

Vectorial representation of velocity:

$$V = \frac{\Delta x}{\Delta t} \quad V = \frac{dx}{dt} \quad \vec{V} = \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$\vec{V} = \frac{\Delta x}{\Delta t}\hat{i} + \frac{\Delta y}{\Delta t}\hat{j}$$



Vectorial representation of Acceleration:

$$a = \frac{\Delta v}{\Delta t} \quad \left| \quad a = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right)$$

$$a = \frac{d^2x}{dt^2}$$

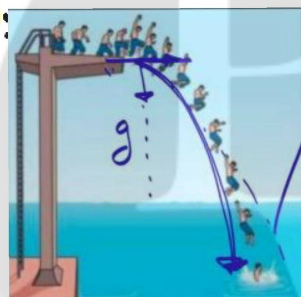
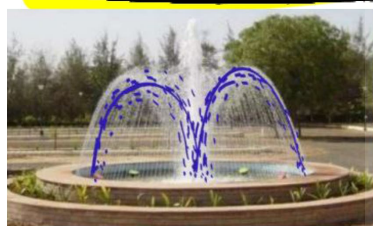
$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

$$\vec{a} = \frac{\Delta v_x}{\Delta t} \hat{i} + \frac{\Delta v_y}{\Delta t} \hat{j}$$

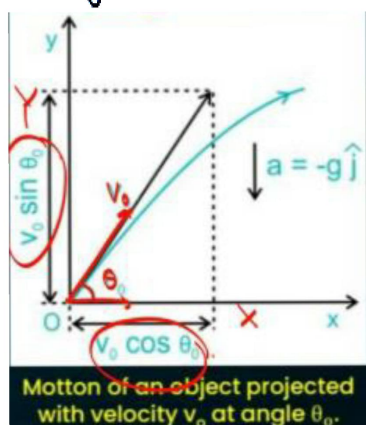
$$\vec{a} = a_x \hat{i} + a_y \hat{j}$$

$$\vec{a} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j}$$

Projectile-Motion:



The motion of an object which is when thrown in air travels or path under the influence of gravity only, neglecting the air resistance, is called the projectile motion of that object.



• Humesha ek planar vector ko oske x-component and y-component main Resolve kardo!

Initial Velocity:

$$V_{0x} = V_0 \cos \theta$$

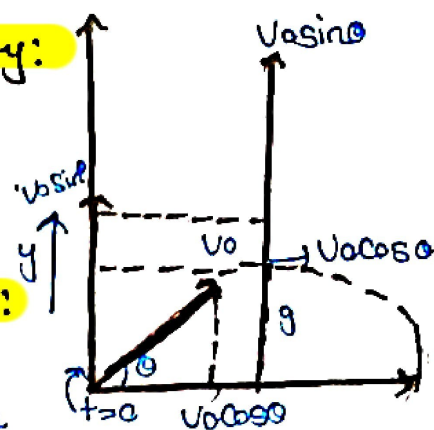
$$V_{0y} = V_0 \sin \theta$$

Final-velocity:

$$V_x = V_0 \cos \theta$$

$$V_y = V_0 \sin \theta + (-g)t$$

$$= V_y = V_0 \sin \theta - gt$$



Displacement: $\{s = ut + \frac{1}{2}at^2\}$

$$x = (V_0 \cos \theta)t + \frac{1}{2}(0)t^2$$

$$\Rightarrow x = (V_0 \cos \theta)t$$

$$y = (V_0 \sin \theta)t + \frac{1}{2}(-g)t^2$$

$$y = (V_0 \sin \theta)t - \frac{1}{2}gt^2$$

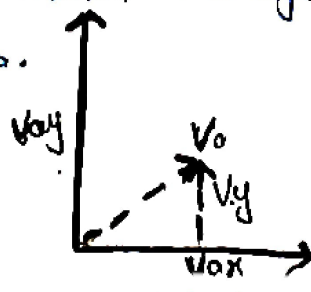
Displacement of the object in the projectile motion in x and y direction at time t =

$$\Rightarrow \left\{ t = \frac{x}{V_0 \cos \theta} \right\} \text{--- (i)}$$

Angle of Projection: The angle made with the horizontal at which the object is projected (or thrown) in the air such that it undergoes projectile motion. The angle is denoted by θ or θ_0 .

$$\tan \theta = \frac{V_{0y}}{V_{0x}}$$

$$\theta = \tan^{-1} \left(\frac{V_{0y}}{V_{0x}} \right)$$



Eder Equation of Path of Projectile:

Equation of Trajectory

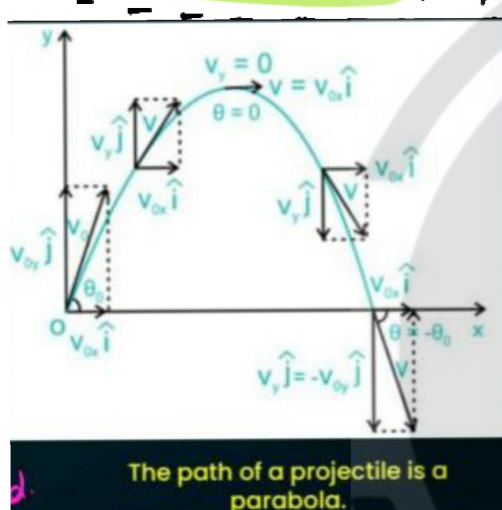
$$y = (\tan \theta)x - \frac{g}{2(v_0 \cos \theta)^2} x^2$$

$$y = (v_0 \sin \theta)t - \frac{1}{2}gt^2$$

Putting eqn (i) in above,

$$y = (v_0 \sin \theta) \left(\frac{x}{v_0 \cos \theta} \right) - \frac{1}{2}g \left(\frac{x}{v_0 \cos \theta} \right)^2$$

$$y = (\tan \theta)x - \frac{g}{2(v_0 \cos \theta)^2} x^2$$



This eqn of trajectory is of the form:

$$y = ax + bx^2$$

eqn of Parabola

∴ we can say that projectile motion follows a parabolic path.

Time of Maximum Height:

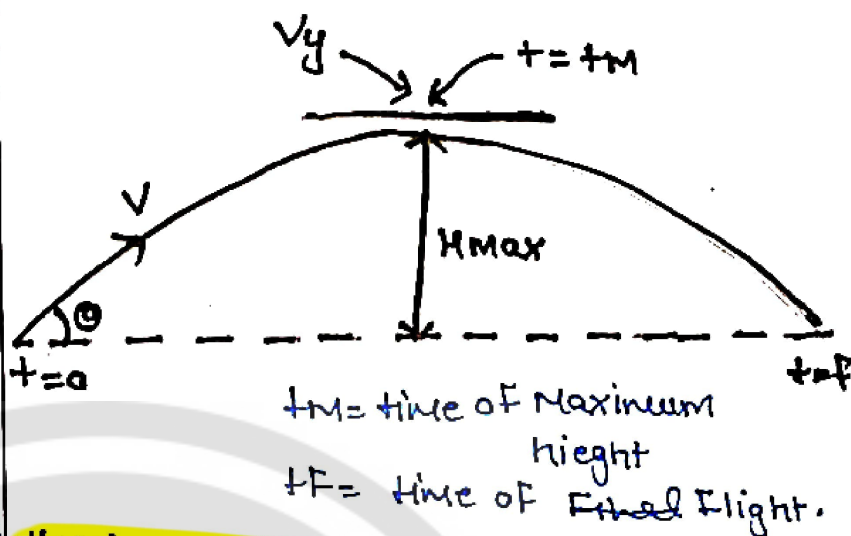
Since, we know that at the velocity of an object under the influence of gravity at the maximum height of its path is zero.

$$v_y = 0$$

$$v_0 \sin \theta - gt = 0$$

$$v_0 \sin \theta - g(tm) = 0$$

$$tm = \frac{v_0 \sin \theta}{g} \leftarrow \text{Time of Max height}$$



Time of Flight:

Time of Flight = 2(Time of Maximum height)

$$\Rightarrow T_F = 2T_M$$

$$\Rightarrow T_F = \frac{2v_0 \sin \theta}{g}$$

Maximum Height of a Projectile:

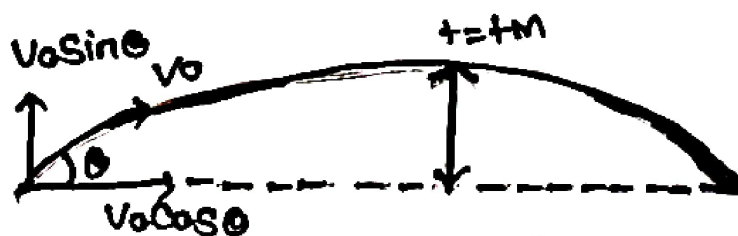
$$S = ut + \frac{1}{2}at^2 \quad \{y\text{-direction}\}$$

$$\Rightarrow H_{max} = (v_0 \sin \theta)(tm) + \frac{1}{2}(-g)(tm)^2$$

$$\Rightarrow H_{max} = (v_0 \sin \theta) \left(\frac{v_0 \sin \theta}{g} \right) - \frac{g}{2} \left(\frac{v_0^2 \sin^2 \theta}{g^2} \right)$$

$$\Rightarrow H_{max} = \frac{v_0^2 \sin^2 \theta}{g} - \frac{v_0^2 \sin^2 \theta}{2g}$$

$$\Rightarrow H_{max} = \frac{v_0^2 \sin^2 \theta}{2g} \leftarrow \text{Maximum height of a projectile.}$$



Horizontal range of a Projectile:

→ x-direction

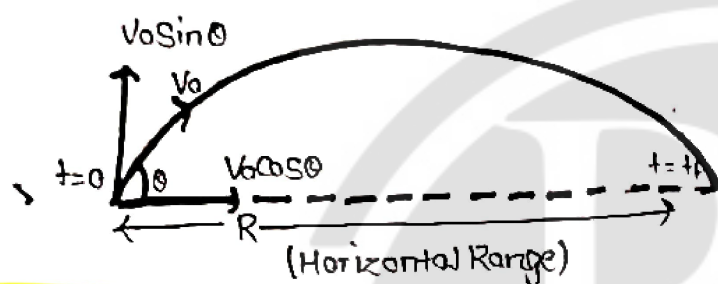
Displacement = (Velocity) (Time)

$$\Rightarrow \text{Range (R)} = (V_0 \cos \theta) (T_f)$$

$$\Rightarrow R = (V_0 \cos \theta) \left(\frac{2 V_0 \sin \theta}{g} \right) = \frac{V_0^2 2 \sin \theta \cos \theta}{g}$$

$$[\sin 2\theta = 2 \sin \theta \cos \theta]$$

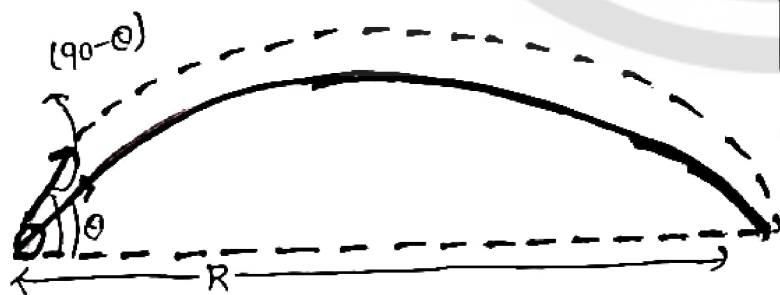
$$\Rightarrow \boxed{R = \frac{V_0^2 \sin 2\theta}{g}}$$



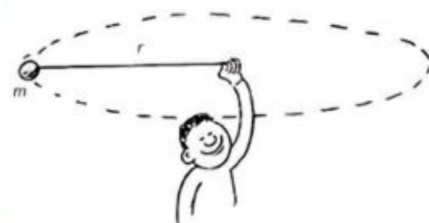
Fact About Projectile Motion:

→ This fact was stated by Galileo.

The angle of Projection of a projectile in two cases where the horizontal Range is same, the sum of the angles is always equal to 90° .

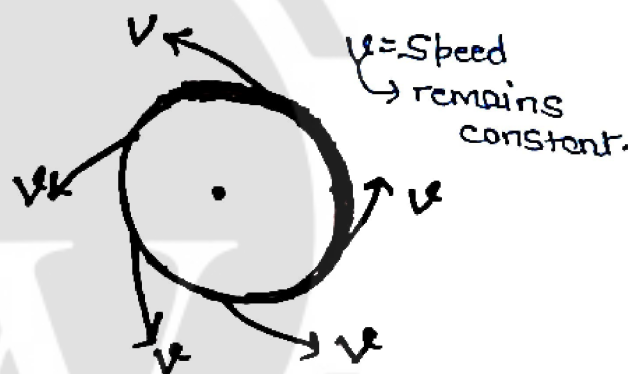


Circular Motion:

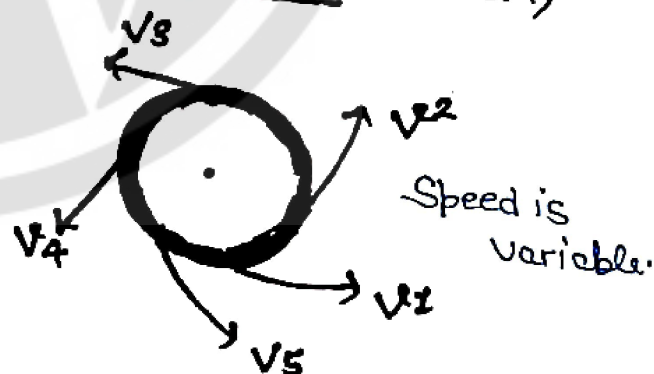


• When a particle moves in a circular path along with a constant speed, it is said to be in uniform circular motion (UCM).

Uniform Circular Motion: (UCM)



Non-Uniform Motion: (NUCM)



Angular Quantities in Circular Motion:

Circular Motion

• Angular displacement
→ (θ)

Linear Motion

• Linear Displacement
→ (y)

• Angular Velocity $\rightarrow$
 $\omega = \frac{d\theta}{dt}$

• Linear Velocity $\rightarrow$
 $v = \frac{dr}{dt}$

• Angular Acceleration $\rightarrow$
 $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

• Linear Acceleration $\rightarrow$
 $a = \frac{dv}{dt} = \frac{d^2r}{dt^2}$

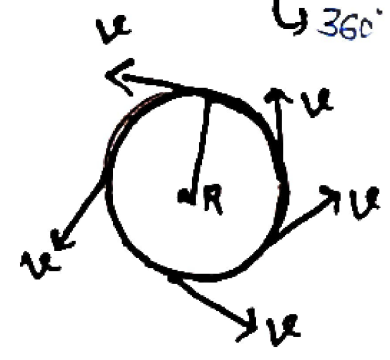
Angular Velocity:

"Omega" $\omega = \frac{d\theta}{dt}$
 Angle covered during circular motion (Radian)
 Time taken (s)
 Angular velocity in rad/s

Relationship btw linear velocity and angular velocity:

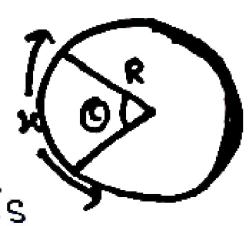
Linear Velocity $= v = \frac{dr}{dt}$
 $\Rightarrow v = \frac{d(R\theta)}{dt} = R \left(\frac{d\theta}{dt} \right) \Rightarrow v = R\omega$
 Linear displacement
 Angular velocity
 Reln btw ω and v

$360^\circ \rightarrow 2\pi$ Radian
 360° in Radian



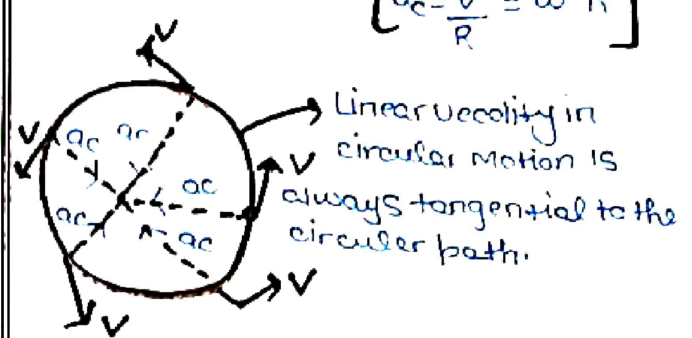
$s = R\theta$

where θ is in Radian.



Centripetal - Acceleration:

$a_c = \frac{v^2}{R} = \omega^2 R$

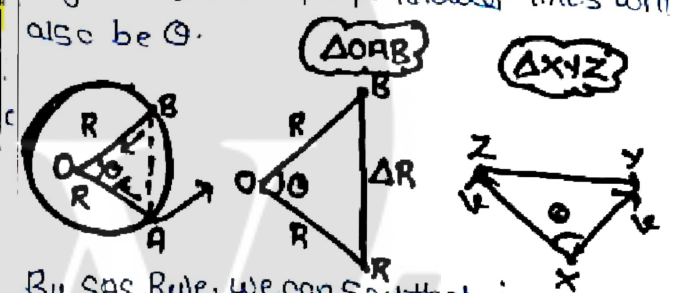


Note: a_c is not constant, but the magnitude of a_c is constant throughout the circular motion

• Acceleration of the object moving in a circular path acting toward the centre of the circle is called centripetal acceleration.

Derivation of Centripetal Acceleration:

If the angle btw two lines is θ , then the angle btw their perpendicular lines will also be θ .



By SAS Rule, we can say that, in $\triangle OAB$ and $\triangle XYZ$,

$\frac{\Delta v}{v} = \frac{\Delta R}{R} \Rightarrow \Delta v = \frac{v \Delta R}{R} = \left(\frac{\Delta v}{\Delta t} \right) = \frac{v}{R} \left(\frac{\Delta R}{\Delta t} \right)$

$\Rightarrow a_c = \omega^2 R$ $\left[a_c = \frac{v^2}{R} \right]$

Centripetal-Force:

Jaha per acc - (chela) Hogi, waha per Force (Guruji) bhi honge. and vice-versa.

Force $F = m a$
 Mass of object Acceleration

In case of U.C.M:-

we have, Force = F_c = Centripetal force
 Mass of the object = m
 Acceleration = a_c = Centripetal Acc

$F_c = m a_c \rightarrow F_c = \frac{mv^2}{R}$
 $F_c = m \omega^2 R$